

Stress Contagion and Tipping Points in Crowd Evacuation Dynamics

*A Mathematical Modeling Study using the
Smooth Social Force Model with Coupled Stress Dynamics*

Research Question

If stress and panic propagate through a crowd like a contagion, is there a critical coupling strength above which the system tips irreversibly into mass panic, and what does this tipping point look like mathematically?

Mathematical Modeling

May 5, 2026

Abstract

We study the propagation of stress and panic in a crowd of evacuating pedestrians by coupling a scalar stress ODE to the Smooth Social Force Model (SSFM). Each agent carries a stress level $\beta_i(t) \in [0, 1]$ that spreads to nearby neighbours and in turn raises the agent's desired speed, connecting panic to the well-known faster-is-slower clogging mechanism. A mean-field reduction yields a single autonomous ODE whose bifurcation structure is fully characterised: a *transcritical bifurcation* at a critical contagion strength $\kappa_c = \mu/(\rho\bar{w})$ separates a globally calm phase from a globally panicked phase. A cubic self-reinforcement extension produces a *saddle-node bifurcation* and bistability, giving rise to *hysteresis*: the crowd tips into panic at a higher threshold than the one required to calm it back down. We derive all equilibria and their stability analytically, relate the critical threshold to crowd density, and outline the companion agent-based simulation that validates the mean-field predictions.

Contents

1	Introduction and Motivation	3
1.1	Goals of This Paper	3
1.2	Paper Outline	3
2	The Smooth Social Force Model (SSFM)	3
2.1	The Full Social Force Model	3
2.2	The Smooth Reduction	4
2.3	Dimensionless Form and the Parameter α	4
3	The Stress-Contagion Augmentation	5
3.1	Individual Stress Variables	5
3.2	Stress Modifies Desired Speed	5
3.3	The Stress Contagion ODE	5
3.4	Why the $(1 - \beta_i)$ Factor is Essential	6
3.5	The Full Coupled System	6
4	Mean-Field Reduction	7
4.1	Assumptions	7
4.2	Derivation	7
5	Equilibrium Analysis and Transcritical Bifurcation	8
5.1	Finding All Equilibria	8
5.2	Stability Analysis	8
5.2.1	Stability of $\beta_0^* = 0$	8
5.2.2	Stability of $\beta_+^* = 1 - \mu/\Lambda$ (when $\Lambda > \mu$)	8
5.3	The Transcritical Bifurcation	9
5.4	Critical Threshold in Terms of Original Parameters	9
5.5	Phase Diagram in (ρ, κ) Space	10
6	Hysteresis: The Saddle-Node Extension	10
6.1	Motivation: Self-Reinforcement of Panic	10
6.2	Equilibria of the Extended Model	10
6.3	Bistability and Saddle-Node Bifurcation	11
7	Coupling Stress to Evacuation Efficiency	12
7.1	The Stress–Speed–Alpha Chain	12
7.2	Evacuation Time as a Function of Mean Stress	13
7.3	Composite Phase Diagram	13
8	Numerical Simulation	13
8.1	Agent-Based Implementation	13
8.2	Planned Numerical Experiments	14
8.2.1	Experiment 1: Validation of Mean-Field Prediction	14
8.2.2	Experiment 2: Hysteresis Loop	14
8.2.3	Experiment 3: Evacuation Time vs. Contagion Strength	14
8.2.4	Experiment 4: Density Dependence of κ_c	14
8.3	Key Macroscopic Observables	15

8.4 Numerical Parameters	15
9 Discussion	15
9.1 Summary of Analytical Results	15
9.2 Limitations and Extensions	16
Notation Summary	17
References	17

1 Introduction and Motivation

Crowd disasters are not caused by force alone. Historical events — the 2010 Love Parade stampede, the 1989 Hillsborough disaster, the 2015 Mina crush — share a common precursor: *a rapid, contagious spread of panic* that transformed an orderly crowd into a chaotic, lethal mass before any physical bottleneck had fully formed. Understanding when and why this transition occurs is a central open problem in crowd science.

The standard Social Force Model (SFM) of Helbing, Farkas, and Vicsek (2000) treats pedestrians as particles subject to a self-driving force, social repulsion, and contact forces. It captures mechanical phenomena such as clogging and lane formation extremely well. What it does *not* natively model is the *psychological dynamics* of the crowd: how fear, observed from a neighbor, raises one’s own anxiety, which in turn modifies behavior and radiates further.

Bellomo et al. (2016) identified this gap explicitly: in their kinetic model, the stress parameter $\beta \in [0, 1]$ is a fixed scalar, but they noted that “ β can evolve in time, hence it should be treated as an additional internal variable. The modelling should include individual modification of β due to interactions. This approach is deemed to model *propagation of the stress conditions*.” Our project answers precisely this call.

1.1 Goals of This Paper

- (1) Formulate a coupled ODE system: the SSFM for agent motion, augmented with a per-agent stress equation with contagion.
- (2) Perform a **mean-field reduction** to a single autonomous scalar ODE.
- (3) Characterise **all equilibria and their stability** analytically.
- (4) Identify the **bifurcation type** (transcritical; saddle-node with hysteresis) and derive the **critical thresholds** in closed form.
- (5) Link the stress dynamics mechanistically back to the **faster-is-slower effect** via the dimensionless parameter $\alpha = A\tau/(mv_0)$.
- (6) Outline the **numerical simulation** strategy for agent-based validation.

1.2 Paper Outline

Section 2 reviews the SSFM and its key dimensionless parameter α . Section 3 introduces the stress-contagion augmentation. Section 4 derives the mean-field reduction. Section 5 performs the full equilibrium and bifurcation analysis. Section 6 extends to the saddle-node / hysteresis case. Section 7 makes the coupling to evacuation efficiency precise. Section 8 outlines the simulation methodology. Section 9 discusses limitations and extensions.

2 The Smooth Social Force Model (SSFM)

2.1 The Full Social Force Model

The standard SFM, introduced by Helbing and Molnár (1995) and extended to crowds by Helbing, Farkas, and Vicsek (2000), governs pedestrian i by:

$$m_i \frac{d\mathbf{v}_i}{dt} = \underbrace{m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}}_{\text{self-driving}} + \underbrace{\sum_{j \neq i} \left[A_i e^{(r_{ij} - d_{ij})/B_i} \mathbf{n}_{ij} + k g(r_{ij} - d_{ij}) \mathbf{n}_{ij} + \kappa g(r_{ij} - d_{ij}) \Delta v_{ji}^t \mathbf{t}_{ij} \right]}_{\text{exponential repulsion + body compression + sliding friction}} + \sum_W \mathbf{f}_{iW}, \quad (1)$$

together with $\dot{\mathbf{x}}_i = \mathbf{v}_i$. The contact function $g(x) = \max(x, 0)$ introduces a Filippov non-smoothness at $d_{ij} = r_{ij}$ that obstructs standard bifurcation analysis.

2.2 The Smooth Reduction

Setting $k = \kappa = 0$ eliminates the contact terms. The **Smooth Social Force Model** (SSFM) is:

$$m \frac{d\mathbf{v}_i}{dt} = \frac{m}{\tau} (v_0(\beta_i) \mathbf{e}_i(\mathbf{x}_i) - \mathbf{v}_i) + A \sum_{j \neq i} e^{(2r - d_{ij})/B} \mathbf{n}_{ij} + \sum_W A_W e^{(r - d_{iW})/B_W} \mathbf{n}_{iW}, \quad (2)$$

where $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ and $\mathbf{n}_{ij} = (\mathbf{x}_i - \mathbf{x}_j)/d_{ij}$. The right-hand side is smooth on the open domain $\mathcal{D} = \{z : d_{ij}(z) > 0 \ \forall i \neq j\}$, which is forward-invariant since the exponential repulsion diverges as $d_{ij} \rightarrow 0^+$.

Remark 2.1: Why smoothness matters

On $\mathcal{D}$, the vector field $F : \mathbb{R}^{4N} \rightarrow \mathbb{R}^{4N}$ is C^∞ . This means the Jacobian DF exists everywhere, linearisation is valid, and the classical Hartman–Grobman theorem, centre manifold theorem, and Hopf/saddle-node/transcritical bifurcation theorems all apply without modification.

2.3 Dimensionless Form and the Parameter α

Introduce natural scales $L_0 = v_0 \tau$, $\tilde{x} = x/(v_0 \tau)$, $\tilde{v} = v/v_0$, $\tilde{t} = t/\tau$. The SSFM becomes:

$$\frac{d\tilde{\mathbf{v}}_i}{d\tilde{t}} = (\mathbf{e}_i - \tilde{\mathbf{v}}_i) + \alpha \sum_{j \neq i} e^{(\tilde{r} - \tilde{d}_{ij})/\tilde{B}} \mathbf{n}_{ij}, \quad (3)$$

where the **repulsion-to-drive ratio** is:

$$\alpha = \frac{A\tau}{mv_0}. \quad (4)$$

Physical Regimes Controlled by α

Regime	Condition	Behavior
Repulsion-dominated	$\alpha \gg 1$	Agents spread out; free flow
Balanced	$\alpha \sim 1$	Moderate crowding
Drive-dominated	$\alpha \ll 1$	High pressure near exit; clogging

Since $\alpha = A\tau/(mv_0)$, *increasing* desired speed v_0 *decreases* α , pushing the crowd toward clogging. This is the mathematical mechanism of the **faster-is-slower effect**.

3 The Stress-Contagion Augmentation

3.1 Individual Stress Variables

We augment the SSFM by assigning to each agent i a **stress level**:

$$\beta_i(t) \in [0, 1], \quad (5)$$

where $\beta_i = 0$ represents a completely calm agent and $\beta_i = 1$ represents an agent in full panic.

3.2 Stress Modifies Desired Speed

Stress directly controls the agent's desired speed via a linear interpolation:

$$v_0(\beta_i) = v_{\min} + \beta_i (v_{\max} - v_{\min}) =: v_{\min} + \beta_i \Delta v, \quad (6)$$

with $v_{\min} \approx 1.0$ m/s (calm walk) and $v_{\max} \approx 4.0$ m/s (full panic run). Since $\alpha_i = A\tau/(mv_0(\beta_i))$, we have:

$$\alpha_i(\beta_i) = \frac{A\tau}{m(v_{\min} + \beta_i \Delta v)}, \quad (7)$$

which is a *monotonically decreasing* function of β_i . Higher stress $\Rightarrow$ higher desired speed $\Rightarrow$ lower $\alpha \Rightarrow$ stronger tendency to clutter and clog.

3.3 The Stress Contagion ODE

The stress level $\beta_i(t)$ evolves according to:

$$\dot{\beta}_i = \underbrace{-\mu \beta_i}_{\text{natural calming}} + \underbrace{\kappa \sum_{j \neq i} w_{ij} \beta_j (1 - \beta_i)}_{\text{contagion from neighbours}} + \underbrace{\sigma_i(t)}_{\text{external shock}}. \quad (8)$$

Parameter Definitions

$\mu > 0$	Natural calming rate. In isolation, agent stress decays exponentially with time constant $1/\mu$.
$\kappa > 0$	Contagion coupling strength. The primary bifurcation parameter. Controls how effectively stress spreads between agents.
w_{ij}	Proximity kernel. Encodes that nearby agents spread stress more effectively: $w_{ij} = e^{-d_{ij}/\ell}$, where $\ell > 0$ is the spatial influence range.
ℓ	Influence range. Typical value $\ell \approx 2\text{--}5$ m.
$(1 - \beta_i)$	Susceptibility factor. A fully panicked agent ($\beta_i = 1$) cannot become more stressed: saturation is built-in. This is the SIR-type nonlinearity.
$\sigma_i(t)$	External shock. Represents a trigger event (e.g. a gunshot, an alarm, a crowd surge) at $t = 0$. Set $\sigma_i(t) = \sigma_0 \delta(t)$ for an instantaneous perturbation.

3.4 Why the $(1 - \beta_i)$ Factor is Essential

The factor $(1 - \beta_i)$ in the contagion term is not merely a convenience; it has three critical consequences:

1. **Boundedness:** $\beta_i \in [0, 1]$ is forward-invariant. If $\beta_i = 1$ then $\dot{\beta}_i = -\mu < 0$; if $\beta_i = 0$ and $\sigma_i = 0$ then $\dot{\beta}_i = 0$.
2. **Nonlinearity:** The equation is quadratic in β , which is what produces the bifurcation. A linear contagion term would give only exponential growth or decay, with no qualitative transition.
3. **Saturation:** At high stress levels, the marginal effect of additional exposure is reduced. This mirrors psychological saturation of fear.

3.5 The Full Coupled System

Collecting everything, the complete model for N agents is the ODE system on $\mathbb{R}^{4N+N} = \mathbb{R}^{5N}$:

$$\dot{\mathbf{x}}_i = \mathbf{v}_i, \tag{9}$$

$$m\dot{\mathbf{v}}_i = \frac{m}{\tau}(v_0(\beta_i)\mathbf{e}_i - \mathbf{v}_i) + A \sum_{j \neq i} e^{(2r-d_{ij})/B} \mathbf{n}_{ij} + \sum_W A_W e^{(r-d_{iW})/B_W} \mathbf{n}_{iW}, \tag{10}$$

$$\dot{\beta}_i = -\mu\beta_i + \kappa \sum_{j \neq i} e^{-d_{ij}/\ell} \beta_j (1 - \beta_i) + \sigma_i(t), \tag{11}$$

for $i = 1, \dots, N$. The state vector is $z = (\mathbf{x}_1, \mathbf{v}_1, \beta_1, \dots, \mathbf{x}_N, \mathbf{v}_N, \beta_N) \in \mathbb{R}^{5N}$.

4 Mean-Field Reduction

The $5N$ -dimensional system (9)–(11) is analytically intractable for large N . We perform a **mean-field reduction** that collapses the stress dynamics to a single scalar ODE.

4.1 Assumptions

- (A1) **Spatial homogeneity.** Agents are distributed approximately uniformly at density ρ (agents per m^2).
- (A2) **Mean-field approximation.** All agents have approximately the same stress level $\beta_i(t) \approx \beta(t)$ at each instant (well-mixed crowd).
- (A3) **Stationary kernel mean.** The spatial distribution changes slowly enough that the kernel mean $\bar{w} = \frac{1}{N-1} \sum_{j \neq i} w_{ij}$ is approximately constant.

Remark 4.1: Validity of mean-field

Assumption (A2) is strongest at early times when the crowd is well-mixed, or in the limit $N \rightarrow \infty$. At late times, spatial heterogeneity (e.g. a panicked cluster near the exit) causes the mean-field to break down. The agent-based simulation (Section 8) captures these effects without approximation.

4.2 Derivation

Under (A1)–(A3), the number of effective neighbours is $N - 1 \approx \rho L^2$ for a room of area L^2 , and:

$$\sum_{j \neq i} w_{ij} \beta_j \approx (N - 1) \bar{w} \beta(t) \approx \rho L^2 \bar{w} \beta(t). \quad (12)$$

Defining the **effective contagion rate**:

$$\Lambda(\rho) = \kappa \rho \bar{w}, \quad (13)$$

the stress equation (11) (ignoring external shocks) reduces to:

$$\dot{\beta} = -\mu \beta + \Lambda(\rho) \beta (1 - \beta). \quad (14)$$

This is the **mean-field stress ODE**. It is a single, autonomous, scalar ODE in $\beta(t) \in [0, 1]$, amenable to complete analytical treatment.

Remark 4.2: Logistic structure

Equation (14) can be rewritten as $\dot{\beta} = (\Lambda - \mu)\beta - \Lambda\beta^2$. This is a **logistic equation** with growth rate $(\Lambda - \mu)$ and carrying capacity $\beta^\infty = 1 - \mu/\Lambda$ (when $\Lambda > \mu$). The logistic ODE is one of the most studied nonlinear ODEs in applied mathematics and its bifurcation structure is classical.

5 Equilibrium Analysis and Transcritical Bifurcation

5.1 Finding All Equilibria

We seek all $\beta^* \in [0, 1]$ satisfying $\dot{\beta} = 0$:

$$-\mu \beta^* + \Lambda \beta^* (1 - \beta^*) = 0 \implies \beta^* [\Lambda(1 - \beta^*) - \mu] = 0. \quad (15)$$

Equation (15) factors immediately into two cases:

Equilibria of the Mean-Field ODE

$$\text{Equilibrium 1 (calm):} \quad \beta_0^* = 0, \quad \text{exists for all } \Lambda, \mu > 0. \quad (16)$$

$$\text{Equilibrium 2 (panicked):} \quad \beta_+^* = 1 - \frac{\mu}{\Lambda}, \quad \text{exists in } [0, 1] \text{ iff } \Lambda \geq \mu. \quad (17)$$

Remark 5.1: Physical interpretation

$\beta_+^* = 0$ when $\Lambda = \mu$ (the two equilibria collide at the origin) and $\beta_+^* \rightarrow 1$ as $\Lambda \rightarrow \infty$ (very strong contagion drives the whole crowd to full panic). The equilibrium crowd panic level grows with contagion strength and shrinks with calming rate — both intuitively correct.

5.2 Stability Analysis

Let $f(\beta) = -\mu\beta + \Lambda\beta(1 - \beta)$. The stability of an equilibrium β^* is determined by the sign of $f'(\beta^*)$:

$$f'(\beta) = \frac{df}{d\beta} = -\mu + \Lambda(1 - 2\beta). \quad (18)$$

5.2.1 Stability of $\beta_0^* = 0$

$$f'(0) = -\mu + \Lambda. \quad (19)$$

- If $\Lambda < \mu$: $f'(0) < 0 \implies \beta_0^* = 0$ is **asymptotically stable**.
- If $\Lambda > \mu$: $f'(0) > 0 \implies \beta_0^* = 0$ is **unstable**.
- If $\Lambda = \mu$: $f'(0) = 0 \implies$ **bifurcation point**.

5.2.2 Stability of $\beta_+^* = 1 - \mu/\Lambda$ (when $\Lambda > \mu$)

$$f'(\beta_+^*) = -\mu + \Lambda \left(1 - 2 \left(1 - \frac{\mu}{\Lambda} \right) \right) = -\mu + \Lambda \left(\frac{2\mu}{\Lambda} - 1 \right) = -\mu + 2\mu - \Lambda = \mu - \Lambda. \quad (20)$$

Since $\Lambda > \mu$, we have $f'(\beta_+^*) = \mu - \Lambda < 0$: β_+^* is **always asymptotically stable** whenever it exists.

5.3 The Transcritical Bifurcation

Theorem 5.1: Transcritical Bifurcation

Let $\Lambda_c = \mu$. The mean-field ODE (14) undergoes a **transcritical bifurcation** at $\Lambda = \Lambda_c$. Specifically:

- For $\Lambda < \Lambda_c$: the calm equilibrium $\beta_0^* = 0$ is the unique attractor in $(0, 1]$. Any initial panic decays to zero.
- At $\Lambda = \Lambda_c$: the calm equilibrium is non-hyperbolic ($f'(0) = 0$); the panicked branch β_+^* collides with it.
- For $\Lambda > \Lambda_c$: $\beta_0^* = 0$ becomes unstable; the panicked equilibrium $\beta_+^* = 1 - \mu/\Lambda$ is the unique attractor. Any initial perturbation $\beta(0) > 0$ leads to sustained collective panic.

The two branches exchange stability at $\Lambda = \Lambda_c$ — the defining signature of a transcritical bifurcation.

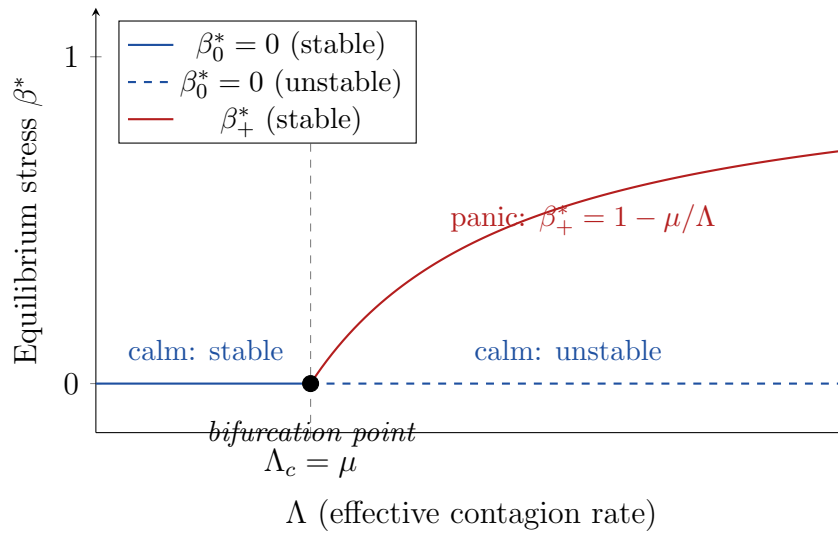


Figure 1: Bifurcation diagram of the mean-field stress ODE (14). Solid lines indicate stable equilibria; dashed lines indicate unstable ones. At $\Lambda = \Lambda_c = \mu$ the calm and panicked branches collide and exchange stability — the hallmark of a transcritical bifurcation.

5.4 Critical Threshold in Terms of Original Parameters

Substituting $\Lambda(\rho) = \kappa\rho\bar{w}$ into $\Lambda_c = \mu$:

$$\kappa_c = \frac{\mu}{\rho\bar{w}}. \quad (21)$$

Dependence of Critical Threshold on Density

$$\kappa_c \propto \frac{1}{\rho}.$$

Denser crowds are more fragile to panic. In a dense crowd (ρ large), even weak contagion (κ small) can exceed the threshold κ_c and trigger a collective panic

cascade. Conversely, a sparse crowd requires much stronger contagion to tip.

5.5 Phase Diagram in (ρ, κ) Space

The boundary curve $\kappa = \kappa_c(\rho) = \mu/(\rho\bar{w})$ divides parameter space into two phases:

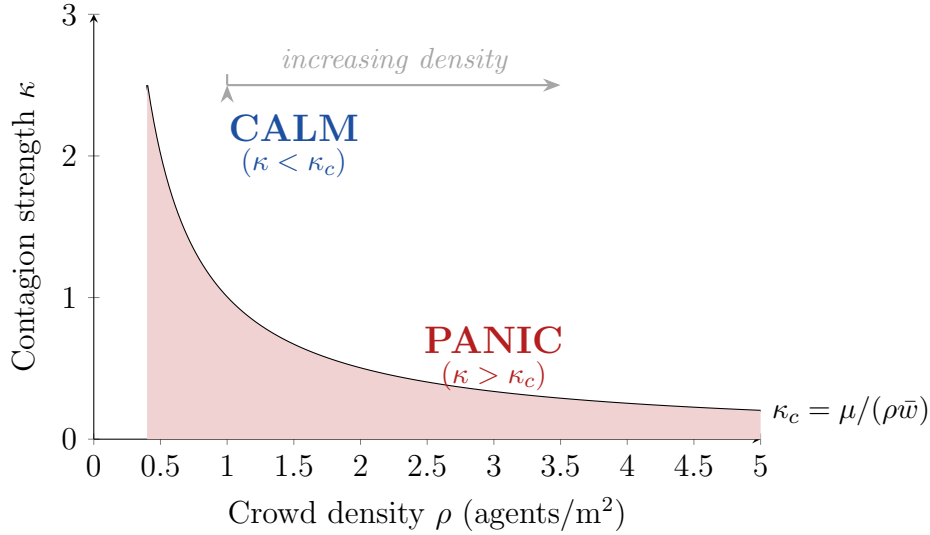


Figure 2: Phase diagram in (ρ, κ) space. The hyperbolic boundary $\kappa_c = \mu/(\rho\bar{w})$ separates the calm phase (above/left) from the panic phase (below/right). At fixed κ , increasing crowd density moves the system rightward, eventually crossing into the panic regime.

6 Hysteresis: The Saddle-Node Extension

The transcritical bifurcation predicts that the transition is *reversible*: reducing κ below κ_c should calm the crowd. This is, however, psychologically unrealistic. Once a crowd has tipped into panic, calming it requires more effort than preventing the initial panic. This phenomenon — **hysteresis** — requires a richer bifurcation structure.

6.1 Motivation: Self-Reinforcement of Panic

In addition to contagion from neighbours, a panicked individual reinforces their own panic through psychological feedback: fear of injury, loss of rational agency, and confirmation bias from perceived threats. We model this by adding a **cubic self-reinforcement** term:

$$\dot{\beta} = -\mu\beta + \Lambda\beta(1-\beta) + \underbrace{\gamma\beta^2(1-\beta)}_{\text{self-reinforcement}}, \quad \gamma > 0. \quad (22)$$

The term $\gamma\beta^2(1-\beta)$ is zero at $\beta = 0$ and $\beta = 1$, positive in between, and has a maximum at $\beta = 2/3$.

6.2 Equilibria of the Extended Model

Setting $\dot{\beta} = 0$ and factoring out β :

$$\beta[-\mu + \Lambda(1 - \beta) + \gamma\beta(1 - \beta)] = 0. \quad (23)$$

Apart from $\beta^* = 0$, the non-trivial equilibria satisfy:

$$\Lambda(1 - \beta) + \gamma\beta(1 - \beta) = \mu \implies (1 - \beta)(\Lambda + \gamma\beta) = \mu. \quad (24)$$

Expanding and rearranging into a quadratic in β :

$$\gamma\beta^2 - (\Lambda + \gamma - \mu)\beta + (\Lambda - \mu) = 0. \quad (25)$$

The discriminant is:

$$\Delta = (\Lambda + \gamma - \mu)^2 - 4\gamma(\Lambda - \mu) = (\Lambda - \mu)^2 + 2(\Lambda - \mu)\gamma + \gamma^2 - 4\gamma(\Lambda - \mu) = (\Lambda - \mu - \gamma)^2 + 4\gamma\mu - 4\gamma(\Lambda - \mu + \mu) \quad (26)$$

which simplifies to:

$$\Delta = (\Lambda - \mu)^2 - 2\gamma(\Lambda - \mu) + \gamma^2 = [(\Lambda - \mu) - \gamma]^2. \quad (27)$$

Remark 6.1: Discriminant check

The discriminant (27) is always ≥ 0 , so real solutions always exist once $\Lambda > \mu$. The critical case $\Delta = 0$ occurs at $\Lambda = \mu + \gamma$, corresponding to a saddle-node point where two branches merge.

The two non-trivial solutions are:

$$\beta_{\pm}^* = \frac{(\Lambda + \gamma - \mu) \pm \sqrt{\Delta}}{2\gamma} = \frac{(\Lambda + \gamma - \mu) \pm |\Lambda - \mu - \gamma|}{2\gamma}. \quad (28)$$

6.3 Bistability and Saddle-Node Bifurcation

The key prediction of the extended model concerns the region $\mu < \Lambda < \mu + \gamma$, where both $\beta_0^* = 0$ and a high-stress equilibrium β_+^* are *stable simultaneously*, separated by an unstable middle branch β_-^* . This is **bistability**.

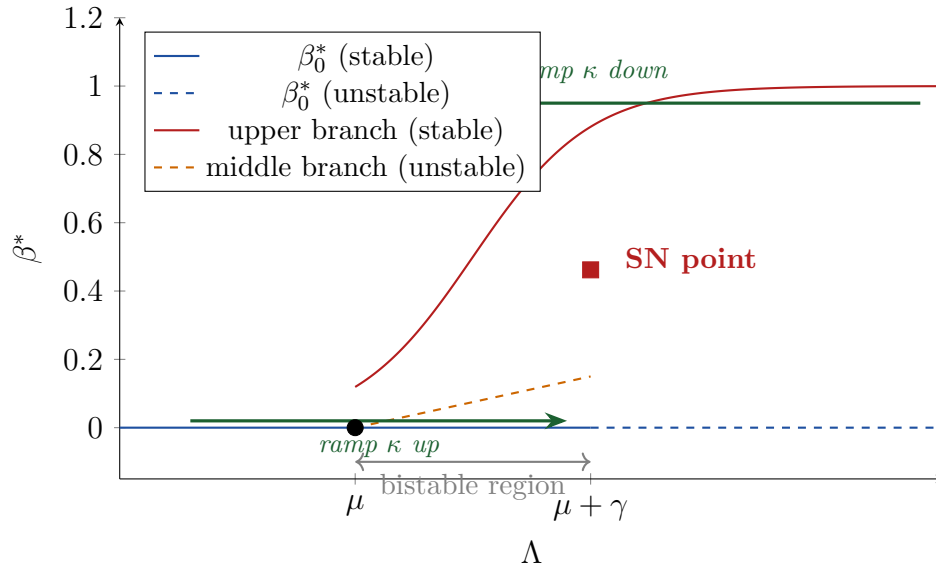


Figure 3: Bifurcation diagram for the extended model with self-reinforcement $\gamma > 0$. In the bistable region $\mu < \Lambda < \mu + \gamma$, both the calm state and the panic state are locally stable. Ramping κ up (green lower arrow), the system stays calm until $\Lambda \approx \mu + \gamma$ where the saddle-node annihilates the calm and unstable branches, and the system *jumps* to the panic branch. Ramping κ back down (green upper arrow), the system remains panicked until $\Lambda \approx \mu$ where the panic branch loses stability. The gap between these two thresholds is the **hysteresis loop**.

Theorem 6.1: Hysteresis

For $\gamma > 0$, the extended model (22) exhibits **irreversible tipping**:

- **Panic threshold** $\Lambda_{\uparrow} = \mu + \gamma$: the calm state loses stability via a saddle-node bifurcation and the system jumps to the panic branch.
- **Recovery threshold** $\Lambda_{\downarrow} = \mu$: the panic branch loses stability and the system returns to calm.
- Since $\Lambda_{\downarrow} < \Lambda_{\uparrow}$, de-panicking requires strictly stronger intervention than panic prevention.

Remark 6.2: Real-world interpretation

Hysteresis explains why simply removing a crowd trigger (e.g. silencing an alarm) may not immediately calm an already-panicked crowd. The crowd is in a meta-stable panicked state; active intervention (calming instructions, clearing exit paths, reducing density) is needed to reduce Λ below $\Lambda_{\downarrow}$.

7 Coupling Stress to Evacuation Efficiency

7.1 The Stress–Speed–Alpha Chain

The full mechanical effect of stress is captured by a chain of dependencies:

$$\kappa \uparrow \implies \beta \uparrow \implies v_0(\beta) \uparrow \implies \alpha(\beta) = \frac{A\tau}{mv_0(\beta)} \downarrow \implies \text{clogging / FIS effect.} \quad (29)$$

Thus **stress contagion and the faster-is-slower effect are mechanistically coupled** through α : contagious panic drives agents to desire higher speeds, which reduces α , pushing the crowd into the clogged regime of the SSFM.

7.2 Evacuation Time as a Function of Mean Stress

In the free-flow regime ($\alpha \gg 1$), the single-agent evacuation theory gives $T_1 = H_{\text{eff}}/v_0(\beta)$. Substituting $v_0(\beta) = v_{\min} + \beta \Delta v$:

$$T_1(\beta) = \frac{H_{\text{eff}}}{v_{\min} + \beta \Delta v}. \quad (30)$$

This is *decreasing* in β in the free-flow regime (faster $\Rightarrow$ shorter evacuation). However, in the clogged regime, the crowd overhead factor $\Gamma = T_{\text{sim}}/T_1$ increases sharply as α decreases (as observed in SSFM simulations). The **actual evacuation time** is:

$$T(\beta) = \Gamma(\alpha(\beta)) \cdot T_1(\beta), \quad (31)$$

where $\Gamma(\alpha)$ is a monotonically decreasing function of α (larger overhead for smaller α). The non-monotone behavior in $T(\beta)$ — the faster-is-slower effect — arises from competition between these two factors.

7.3 Composite Phase Diagram

Combining the panic threshold Λ_c with the clogging threshold $\alpha^* = \alpha_c(\rho)$ from the SSFM yields a **composite phase diagram** in (ρ, κ) space with three regions:

Region	Condition	Behavior
Safe	$\kappa < \kappa_c$ and $\alpha > \alpha_c$	Calm, free flow, short T
Chaotic	$\kappa > \kappa_c$ and $\alpha < \alpha_c$	Panicked <i>and</i> clogged, catastrophic T
Transitional	$\kappa \approx \kappa_c$ or $\alpha \approx \alpha_c$	Metastable; sensitive to perturbations

8 Numerical Simulation

8.1 Agent-Based Implementation

The full $5N$ -dimensional system (9)–(11) is integrated numerically using an explicit **Euler** or **RK4** scheme. The simulation loop at each timestep Δt :

1. **Compute distances** d_{ij} and directions $\mathbf{n}_{ij}$ for all pairs.
2. **Update velocities** $\mathbf{v}_i$ via SSFM forces with $v_0(\beta_i)$.
3. **Update positions** $\mathbf{x}_i = \mathbf{x}_i + \mathbf{v}_i \Delta t$.
4. **Update stresses** β_i via the contagion ODE (11).

5. **Remove** agents that have crossed the exit boundary.

6. **Record** $(\mathbf{x}_i, \mathbf{v}_i, \beta_i)$ and macroscopic quantities.

8.2 Planned Numerical Experiments

8.2.1 Experiment 1: Validation of Mean-Field Prediction

Setup: $N = 100$ agents in a 10×10 m room, uniform initial positions, $\beta_i(0) = \beta_0 \ll 1$ for all i (small initial perturbation), no evacuation (static positions) to isolate the stress dynamics.

Sweep: Vary κ from 0 to $3\kappa_c$. For each κ , run to steady state and record $\bar{\beta}^\infty = \lim_{t \rightarrow \infty} \bar{\beta}(t)$.

Expected outcome: $\bar{\beta}^\infty \approx 0$ for $\kappa < \kappa_c$; $\bar{\beta}^\infty \approx 1 - \mu/\Lambda(\rho)$ for $\kappa > \kappa_c$. The jump at κ_c reproduces Figure 1.

8.2.2 Experiment 2: Hysteresis Loop

Setup: Same room, $N = 100$. Start with $\kappa = 0.1\kappa_c$ (calm phase), slowly ramp κ up to $3\kappa_c$ over time T_{ramp} , then ramp it back down.

Expected outcome: Forward path stays calm until $\kappa \approx \kappa_c + \gamma/(\rho\bar{w})$ where the crowd jumps to panic. Backward path stays panicked until $\kappa \approx \kappa_c$. The gap between these two is the hysteresis loop predicted by Theorem 6.1.

8.2.3 Experiment 3: Evacuation Time vs. Contagion Strength

Setup: $N = 50$ agents, exit of width $w = 1$ m on one wall. Agents start in a circular cluster at the room centre. At $t = 0$, a fraction $p_0 = 0.1$ of agents are initialised with $\beta_i = 0.8$ (seed panickers); all others start at $\beta_i = 0$.

Sweep: Vary κ from 0 to $3\kappa_c$. Record evacuation time T .

Expected outcome: $T(\kappa)$ first *decreases* slightly (mild panic helps agents move faster) then *increases* sharply once $\kappa > \kappa_c$ and clogging kicks in. This produces the coupled panic-FIS curve.

8.2.4 Experiment 4: Density Dependence of κ_c

Setup: Vary N from 20 to 200 in a fixed 10×10 m room (varying ρ). For each ρ , identify $\kappa_c(\rho)$ numerically (the value of κ at which $\bar{\beta}^\infty$ transitions from ≈ 0 to ≈ 1).

Expected outcome: $\kappa_c \propto 1/\rho$, confirming equation (21).

8.3 Key Macroscopic Observables

Observable	Formula	Interpretation
Mean stress	$\bar{\beta}(t) = \frac{1}{N} \sum_i \beta_i(t)$	Global panic level
Panic fraction	$p(t) = \frac{1}{N} \#\{i : \beta_i > 0.5\}$	Fraction of panicked agents
Mean speed	$\bar{v}(t) = \frac{1}{N} \sum_i \ \mathbf{v}_i(t)\ $	Collective speed
Flow rate	$q(t) = \text{exits per unit time}$	Throughput
Evacuation time	$T = \text{time when last agent exits}$	Safety metric
Crowd pressure	$P = \rho \bar{v}^2$	Crush risk indicator

8.4 Numerical Parameters

Symbol	Meaning	Value	Units
m	Agent mass	80	kg
τ	Relaxation time	0.5	s
r	Agent radius	0.25	m
A	Repulsion strength	2000	N
B	Repulsion range	0.08	m
$v_{\min}$	Calm desired speed	1.0	m/s
$v_{\max}$	Panic desired speed	4.0	m/s
μ	Calming rate	0.3	s ⁻¹
κ	Contagion strength	varied	s ⁻¹
ℓ	Influence range	3.0	m
γ	Self-reinforcement	0.5	s ⁻¹
Δt	Timestep	0.02	s

9 Discussion

9.1 Summary of Analytical Results

1. The mean-field stress ODE (14) has a **transcritical bifurcation** at $\kappa_c = \mu/(\rho\bar{w})$, separating a globally calm phase from a globally panicked phase.
2. The critical threshold **decreases with density**: $\kappa_c \propto 1/\rho$. Dense crowds are intrinsically more fragile.
3. Stress drives the SSFM's α parameter downward through the chain (29), mechanistically linking panic to the faster-is-slower clogging effect.
4. A cubic self-reinforcement term $\gamma > 0$ produces a **saddle-node bifurcation** and **hysteresis**: once the crowd tips into panic, calming requires reducing Λ below μ , which is strictly less than the panic threshold $\mu + \gamma$.

9.2 Limitations and Extensions

Limitations

- **Mean-field approximation** ignores spatial heterogeneity (panic clusters near the exit, calm pockets at the back of the room).
- **Homogeneous agents:** real crowds contain children, elderly, and mobility-impaired individuals with different $v_{\min}$, $v_{\max}$, and susceptibility to contagion.
- **No social groups:** panicking groups may resist contagion internally while amplifying it externally.
- **Static contagion kernel:** in a moving crowd, d_{ij} changes rapidly; the assumption of constant $\bar{w}$ may fail late in the evacuation.
- **No information propagation:** in reality, visual cues, noise, and announcements also drive panic.

Extensions for Further Work

- **Spatially resolved stress field:** replace the mean-field with a PDE for $\beta(\mathbf{x}, t)$ coupled to the crowd density $\rho(\mathbf{x}, t)$ — a reaction-diffusion system.
- **Heterogeneous susceptibility:** assign each agent a susceptibility $s_i \in [0, 1]$ sampled from a distribution.
- **Information coupling:** add an “alarm” field that diffuses through the room and raises σ_i .
- **Democratic vs. rioter subdivision:** following Bellomo et al. (2016), subdivide agents into calm majority and panic-seed minority, with transitions modelled as a biological mutation process.

Notation Summary

Symbol	Meaning	Eq.
$\mathbf{x}_i, \mathbf{v}_i$	Position, velocity of agent i	(9)
$\beta_i(t)$	Stress level of agent i , $\in [0, 1]$	(5)
$v_0(\beta_i)$	Stress-dependent desired speed	(6)
$\alpha(\beta_i)$	Repulsion-to-drive ratio	(7)
$\mathbf{e}_i$	Desired direction (unit vector toward exit)	(2)
d_{ij}	Distance between agents i and j	
$\mathbf{n}_{ij}$	Unit vector from j to i	
μ	Natural calming rate	(8)
κ	Contagion coupling strength	(8)
$w_{ij} = e^{-d_{ij}/\ell}$	Proximity kernel	(8)
$\Lambda(\rho) = \kappa\rho\bar{w}$	Effective contagion rate	(13)
$\Lambda_c = \mu$	Critical contagion rate	(21)
$\kappa_c = \mu/(\rho\bar{w})$	Critical coupling strength	(21)
γ	Self-reinforcement coefficient	(22)
$\beta_0^* = 0$	Calm equilibrium	(16)
$\beta_+^* = 1 - \mu/\Lambda$	Panicked equilibrium	(17)
$\sigma_i(t)$	External shock forcing	(8)
$T(\beta)$	Evacuation time	(31)

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